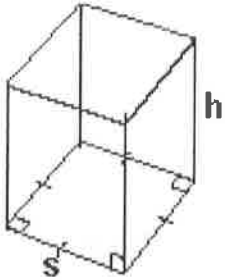


Part 3: Maximize the Volume of a Square-Based Prism

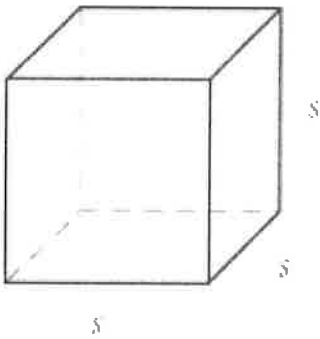
Square-based prism



$$\text{Volume} = s^2 h$$

$$\text{Surface Area} = 2s^2 + 4sh$$

Cube



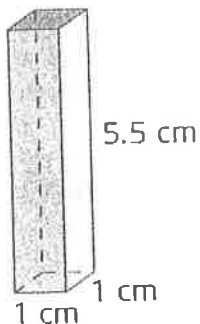
$$\text{Volume} = s^3$$

$$\text{Surface Area} = 6s^2$$

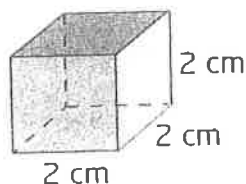
For a given surface area, it is usually optimal to maximize the volume.

Investigation 1: Each of the square-based prisms has a surface area of 24 cm^2 .

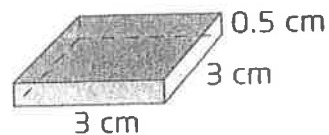
Prism 1



Prism 2



Prism 3



a) Calculate the area of the base and the volume of each prism. Record your data in the following table:

Prism Number	Length of base	Area of base	Surface Area	Height	Volume
1	1	1	24	5.5	5.5
2	2	4	24	2	8
3	3	9	24	0.5	4.5

b) Which square-based prism has the maximum volume? Describe the shape of this prism compared to the others.

Prism 2 has max volume for the given surface area.

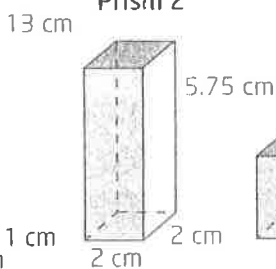
The prism is a cube. $s=h$.

Investigation 2: Each of the square-based prisms has a surface area of 54cm^2 .

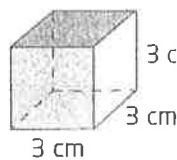
Prism 1



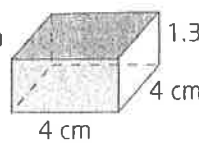
Prism 2



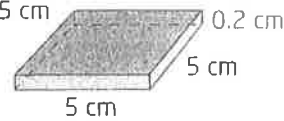
Prism 3



Prism 4



Prism 5



a) Predict the dimensions of the square-based prism with maximum volume if the surface area is 54cm^2 .

Prism 3 will have the max volume for the given surface area because it is a cube.

b) Test your prediction by completing the following table.

Prism Number	Length of base	Area of base	Surface Area	Height	Volume
1	1	1	54	13	13
2	2	4	54	5.75	23
3	3	9	54	3	27
4	4	16	54	1.375	22
5	5	25	54	0.2	5

Results: The maximum volume for a given surface area of a square-based prism always occurs when the prism is a cube.

Example 1: Determine the dimensions of the square-based prism with maximum volume that can be formed using 5400 cm^2 of cardboard.

Solution:

We know that a cube will have the maximum volume

Surface Area of a Cube $= 6s^2$ where s represents the side length

To calculate the side length of the cube given the surface area you must rearrange the formula to isolate s :

$$s = \sqrt{\frac{\text{Surface Area}}{6}}$$

Therefore, if the Surface Area is 5400 cm^2 , $s = \sqrt{\frac{5400}{6}} = 30\text{cm}$. This means that the dimensions of the square-based prism with maximum volume are **30cm x 30cm x 30cm**.

Apply What You Learned

1. Determine the dimensions of the square-based prism with maximum volume for each surface area.

a) 150 cm^2

$$\begin{aligned} s &= \sqrt{\frac{SA}{6}} \\ &= \sqrt{\frac{150}{6}} \\ &= 5 \end{aligned}$$

The dimensions are $5\text{cm} \times 5\text{cm} \times 5\text{cm}$

b) 2400 m^2

$$\begin{aligned} s &= \sqrt{\frac{SA}{6}} \\ &= \sqrt{\frac{2400}{6}} \\ &= 20 \end{aligned}$$

The dimensions are $20\text{m} \times 20\text{m} \times 20\text{m}$

c) 750 cm^2

$$\begin{aligned} s &= \sqrt{\frac{SA}{6}} \\ &= \sqrt{\frac{750}{6}} \\ &= 11.2 \end{aligned}$$

The dimensions are $11.2\text{cm} \times 11.2\text{cm} \times 11.2\text{cm}$

2. Determine the volume of each prism in question 1, to the nearest cubic unit.

a)
$$\begin{aligned} V &= s^3 \\ &= (5)^3 \\ &= 125 \text{ cm}^3 \end{aligned}$$

b)
$$\begin{aligned} V &= s^3 \\ &= (20)^3 \\ &= 8000 \text{ m}^3 \end{aligned}$$

c)
$$\begin{aligned} V &= s^3 \\ &= (11.2)^3 \\ &= 1404.9 \text{ cm}^3 \end{aligned}$$